Duetto ROI Calculator — Metrics & Calculations Reference

This document describes every metric displayed in the ROI calculator, the inputs it depends on, and the exact formula used to derive it.

1. Input Parameters

All monetary inputs are collected in their native currency and converted to the selected output currency before any calculation runs.

Input	Description	Notes
`totalRooms`	Total number of sellable rooms	—
`currentADR`	Current Average Daily Rate	In property currency
`currentOccupancy`	Current occupancy (0–1)	e.g. 0.72 = 72%
`yieldablePercent`	Fraction of room nights the RMS can optimize	Defaults to `1 − groupBusinessPercent`
`groupBusinessPercent`	Fraction of room nights sold as group blocks	Not optimized by open pricing
`hoursPerWeekManual`	Hours/week spent on manual pricing	Used for productivity metric only
`annualRMSystemsCost`	Annual cost of rate shoppers, pricing tools	In property currency
`annualConsultingCost`	Annual outsourced RM / consulting fees	In property currency
`subscriptionCost`	Annual Duetto subscription	In Duetto currency; 0 = auto-estimate
`implementationFee`	One-time go-live fee	In Duetto currency; Year 1 only
`ohipConnectivityFee`	Annual OHIP fee (Opera Cloud PMS only)	In Duetto currency
`initialContractYears`	Length of initial contract term (1–5)	Affects 5-year cost schedule
`revparUpliftPercent`	Scenario assumption: expected RevPAR uplift	Set per scenario (conservative / moderate / aggressive)
`marketGrowthRate`	Annual market growth assumption	Applied in Years 2–5

2. Currency Normalization

All monetary values are converted to the selected output currency before calculations run.

Exchange rate convention: `1 USD = N units` (e.g. MXN: 17.15 means 1 USD = 17.15 MXN).

```
outputValue = (inputValue / rate[inputCurrency]) × rate[outputCurrency]
```

- Performance metrics (ADR, RM costs) convert from `inputs.currency`
- Duetto investment fields convert from `inputs.duettoCurrency`
- Guard: if a rate is 0 or non-finite, it defaults to 1 (no conversion) to prevent divide-by-zero

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3. Baseline Metrics

Current RevPAR

```
currentRevPAR = currentADR × currentOccupancy
```

Annual Available Room Nights

```
annualRooms = totalRooms × 365
```

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4. RevPAR Uplift Model

The model applies a single `revparUpliftPercent` assumption to yieldable inventory, then decomposes it into ADR and occupancy components.

ADR share of RevPAR uplift: 60% (open pricing is primarily rate-driven)

Occupancy share: 40% (demand intelligence)

Yieldable-Segment New Metrics

```
yieldableNewRevPAR = currentRevPAR × (1 + revparUpliftPercent)

yieldableNewADR    = currentADR × (1 + revparUpliftPercent × 0.60)

yieldableNewOcc    = MIN(0.98, yieldableNewRevPAR / yieldableNewADR)
                     [falls back to currentOccupancy if yieldableNewADR = 0]
```

Hotel-Wide Blended Metrics

The uplift on yieldable segments is scaled back to the full room inventory. Non-yieldable nights (group blocks, contracted) remain unchanged.

```
newRevPAR    = currentRevPAR + (yieldableNewRevPAR - currentRevPAR) × yieldablePercent
newADR       = currentADR    + (yieldableNewADR    - currentADR)    × yieldablePercent
newOccupancy = MIN(0.98, newRevPAR / newADR)
```

Why the displayed RevPAR improvement ≠ the scenario uplift input:
The scenario uplift (e.g. 14%) applies only to yieldable inventory. Hotel-wide improvement = `uplift × yieldablePercent`. At 90% yieldable, a 14% input produces a 12.6% hotel-wide improvement.

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5. Annual Revenue Impact

Incremental ADR Revenue

Revenue gained from the rate increase on yieldable rooms at current occupancy levels.

```
incrementalADRRevenue = (yieldableNewADR - currentADR)
                      × currentOccupancy
                      × annualRooms
                      × yieldablePercent
```

Incremental Occupancy Revenue

Revenue from additional room nights filled at the new higher ADR.

```
incrementalOccRevenue = yieldableNewADR
                      × (yieldableNewOcc - currentOccupancy)
                      × annualRooms
                      × yieldablePercent
```

Combined RevPAR Uplift (primary revenue figure)

This is the single combined annual revenue gain used in all summaries.

```
combinedRevPARUplift = (yieldableNewRevPAR - currentRevPAR)
                     × annualRooms
                     × yieldablePercent

annualIncrementalRoomRevenue = combinedRevPARUplift
```

Note: `incrementalADRRevenue + incrementalOccRevenue ≈ combinedRevPARUplift` — minor differences arise because ADR and occupancy components are cross-effects of the same RevPAR uplift. The combined figure is definitive.

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6. Cost Savings

Systems Cost Savings

Direct replacement of existing RM tools and outsourced services that Duetto consolidates.

```
systemsCostSavings = annualRMSystemsCost + annualConsultingCost
```

Strategic Time Reclaimed (productivity metric — not monetized)

```
hoursSavedPerWeek = MIN(hoursPerWeekManual × 0.50, 20)
```

Represents up to 50% of manual pricing time freed. Capped at 20 hrs/week. Shown as a qualitative benefit, not added to financial totals.

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7. Annual Financial Impact

The headline number shown throughout the calculator.

```
totalAnnualImpact = annualIncrementalRoomRevenue + systemsCostSavings
```

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8. Duetto Investment

Annual Recurring Cost

```
duettoAnnualCost = subscriptionCost                      [if manually entered]
                 + ohipConnectivityFee                   [if PMS is Opera Cloud]

                OR

duettoAnnualCost = estimateDuettoCost(totalRooms)         [if subscriptionCost = 0]
                 + ohipConnectivityFee
```

Auto-Estimate Tiers (USD, used when no manual subscription is entered)

Room Count	Estimated Annual Cost
< 100	$18,000
100–199	$28,000
200–349	$42,000
350–499	$58,000
500–749	$75,000
750–999	$95,000
1,000+	$120,000

> Auto-estimates are always in USD and converted to the output currency using the configured exchange rate.

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9. RMS Effectiveness Schedule (Year 1 Learning Curve)

Months 1–2 generate no revenue (implementation / go-live period, ~6–8 weeks). Months 3–12 ramp progressively as the team adopts Duetto.

Default schedule:

Month	1	2	3	4	5	6	7	8	9	10	11	12
Effectiveness	0%	0%	40%	50%	60%	70%	75%	80%	85%	90%	95%	100%

These percentages are fully configurable in Tab 2 → ROI Projection.

Monthly incremental revenue (Year 1):

```
monthlyRevenue[m] = effectiveness[m] × (newRevPAR - currentRevPAR) × totalRooms × 30
```

Year 1 effective ramp factor (used in 5-year projection):

```
year1Ramp = SUM(effectiveness[1..12]) / 12
```

With defaults: `(0+0+40+50+60+70+75+80+85+90+95+100)/12 = 62.1%`

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10. ROI Metrics

ROI Multiple

```
roiMultiple = totalAnnualImpact / duettoAnnualCost
```

Net ROI Percent

```
netROIPercent = ((totalAnnualImpact - duettoAnnualCost) / duettoAnnualCost) × 100
```

Payback Period (months)

Simulated month-by-month. The implementation fee is treated as an upfront cost at contract signing (month 0). Subscription accrues monthly. Revenue accrues according to the RMS effectiveness schedule.

```
cumulativeCost[0]    = implementationFee                    ← at signing
cumulativeCost[m]    = cumulativeCost[m-1] + (duettoAnnualCost / 12)

monthlyBaseImpact    = combinedRevPARUplift / 12
cumulativeRevenue[m] = cumulativeRevenue[m-1] + effectiveness[m] × monthlyBaseImpact

paybackMonth = first m where cumulativeRevenue[m] ≥ cumulativeCost[m]
```

If break-even is not reached within the 12-month effectiveness schedule, the payback is projected beyond Year 1 assuming 100% effectiveness from Month 13 onwards:

```
deficit      = cumulativeCost[12] - cumulativeRevenue[12]
netMonthly   = monthlyBaseImpact - (duettoAnnualCost / 12)
paybackMonths = 12 + CEIL(deficit / netMonthly)
```

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11. 5-Year Value Creation

Per-Year Duetto Investment Schedule

```
Year 1:            duettoAnnualCost + implementationFee
Years 2..N:        duettoAnnualCost                           (N = initialContractYears)
Years N+1..5:      duettoAnnualCost × 1.05^(year − N)         (5% annual escalation post-contract)
```

Per-Year Revenue and Impact

```
rampFactor[year]      = year1Ramp          (Year 1, from effectiveness schedule)
                      = 1.0                (Years 2–5)

marketMultiplier[year] = (1 + marketGrowthRate)^(year − 1)

incrementalRevenue[year] = baseProjection.totalIncrementalRevenue
                         × marketMultiplier[year]
                         × rampFactor[year]

costSavings[year]       = baseProjection.totalCostSavings
                         × marketMultiplier[year]
                         × rampFactor[year]

totalImpact[year]       = incrementalRevenue[year] + costSavings[year]
```

Net Benefit and Cumulative

```
netBenefit[year]         = totalImpact[year] − duettoInvestment[year]
cumulativeNetBenefit[y]  = SUM(netBenefit[1..y])
```

Per-Year ROI %

```
roiPercent[year] = ((totalImpact[year] − duettoInvestment[year]) / duettoInvestment[year]) × 100
```

5-Year Aggregate Metrics

```
totalFiveYearImpact      = SUM(totalImpact[1..5])
totalFiveYearNet         = SUM(netBenefit[1..5])
totalFiveYearInvestment  = SUM(duettoInvestment[1..5])
fiveYearROIMultiple      = totalFiveYearImpact / totalFiveYearInvestment
```

Property Valuation Impact

```
incrementalNOI   = totalImpact[Year 5] × 0.85      ← 85% flow-through to NOI
valuationImpact  = incrementalNOI / capRate
```

The cap rate defaults to 7.5% and is configurable in the Financial Assumptions panel.

12. Portfolio Mode Aggregation

When 2+ properties are entered, all inputs are aggregated into a single blended set before calculations run.

Room-count weighted average (for rate and mix metrics):

```
blendedADR         = SUM(property.currentADR × property.totalRooms) / totalRooms
blendedOccupancy   = SUM(property.currentOccupancy × property.totalRooms) / totalRooms
blendedGroupPct    = SUM(property.groupBusinessPercent × property.totalRooms) / totalRooms
blendedYieldable   = SUM(property.yieldablePercent × property.totalRooms) / totalRooms
```

Summed (for cost metrics):

```
totalRooms         = SUM(property.totalRooms)
annualRMSystemsCost = SUM(property.annualRMSystemsCost)
annualConsultingCost = SUM(property.annualConsultingCost)
duettoAnnualCost   = SUM(property.duettoAnnualCost OR estimateDuettoCost(property.totalRooms))
```

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13. Market Segment Benchmark (CoStar)

When a CoStar Hospitality Market Report is uploaded, the calculator computes the year-over-year change for the property's hotel class segment.

Segment classes: Luxury & Upper Upscale · Upscale & Upper Midscale · Midscale & Economy

Segment YoY Calculation

Uses the two most recent non-forecast years from the segment's historical data.

```
revparPct = ((currentYear.revpar − priorYear.revpar) / priorYear.revpar) × 100
adrPct    = ((currentYear.adr − priorYear.adr) / priorYear.adr) × 100
occDelta  = currentYear.occupancy − priorYear.occupancy          ← in percentage points
```

Market Outperformance (Executive Summary)

```
outperformancePP = projectedRevPARImprovement% − segmentYoY.revparPct
```

Positive values mean the property's projected growth with Duetto exceeds the market segment's recent historical trend.

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14. Scenario Defaults

Scenario	RevPAR Uplift	Market Growth Rate
Conservative	5%	2.0% / year
Moderate	8%	2.5% / year
Aggressive	12%	3.0% / year

All values are fully overridable. Market signals from Duetto client hotel data or CoStar YoY data can be synced directly to the scenario assumptions.

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15. What Is and Is Not Included

Revenue Source	Included	Notes
RevPAR uplift (ADR + occupancy) on yieldable inventory	✅	Primary revenue driver
Systems cost savings (rate shoppers, RM tools, consulting)	✅	Direct cost replacement
Group revenue uplift	❌	Removed — group contracts are typically pre-negotiated
OTA commission savings / distribution shift	❌	Removed — depends on channel strategy outside Duetto scope
Labor cost monetization	❌	Time saved is shown as a productivity metric only, not added to financial totals

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